BEING LOCALLY OF FINITE TYPE CAN BE CHECKED AFTER PRECOMPOSING WITH A FLAT PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The definition of a morphism of algebraic stacks being locally of finite type through smooth covers by algebraic spaces. The author's section on morphisms of finite type, in which the counted statement occurs. Reduces the assertion about stacks to an assertion about morphisms of algebraic spaces.
- The descent of the property of being locally of finite type along a surjective flat morphism of finite presentation. Invoked at the end of the counted proof (source lines 3601-3604) as a cross-chapter pointer. Provides the statement for algebraic spaces to which the proof reduces.

Source reference. Original reference spaces-descent-lemma-locally-finite-type-fppf-local-source: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3604–3604. Navigation pointer only: the counted text refers to the Stacks item spaces-descent-lemma-locally-finite-type-fppf-local-source, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X} \to \mathcal{Y} \to \mathcal{Z}$ be morphisms of algebraic stacks. Assume $\mathcal{X} \to \mathcal{Z}$ is locally of finite type and that $\mathcal{X} \to \mathcal{Y}$ is representable by algebraic spaces, surjective, flat, and locally of finite presentation. Then $\mathcal{Y} \to \mathcal{Z}$ is locally of finite type.*

Proof. Choose an algebraic space W and a surjective smooth morphism $W \to \mathcal{Z}$. Choose an algebraic space V and a surjective smooth morphism $V \to W \times_{\mathcal{Z}} \mathcal{Y}$. Set $U = V \times_{\mathcal{Y}} \mathcal{X}$ which is an algebraic space. We know that $U \to V$ is surjective, flat, and locally of finite presentation and that $U \to W$ is locally of finite type.

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Hence the lemma reduces to the case of morphisms of algebraic spaces. The case of morphisms of algebraic spaces is Descent on Spaces, Lemma spaces-descent-lemma-locally-finite-type-fppf-local-source. □